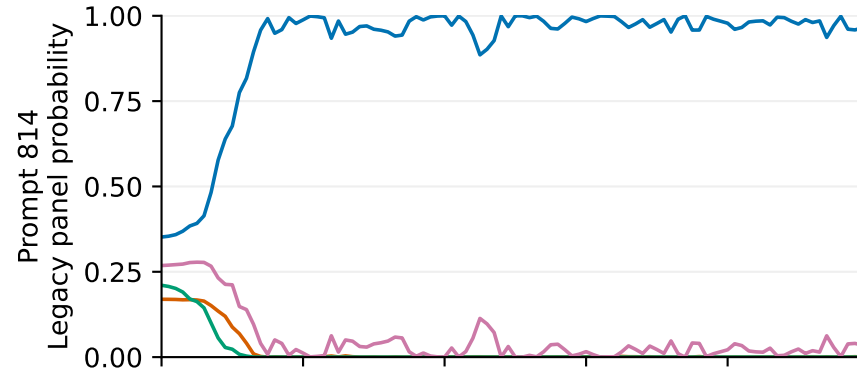


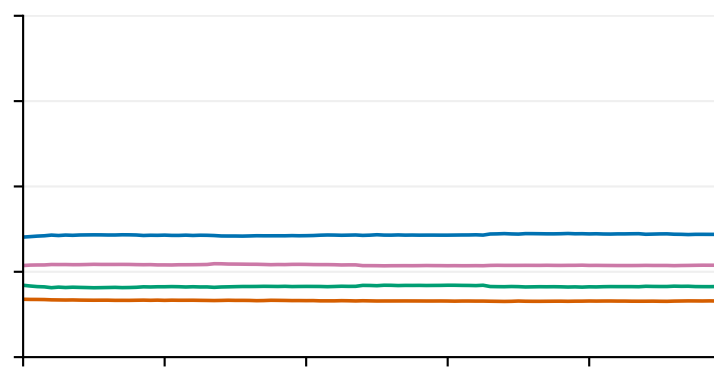
IPO | Original cyclic experiments

Legacy definition: $\pi_i = \text{softmax}(\text{response-token-average log probability})$; not sequence-sum π_i

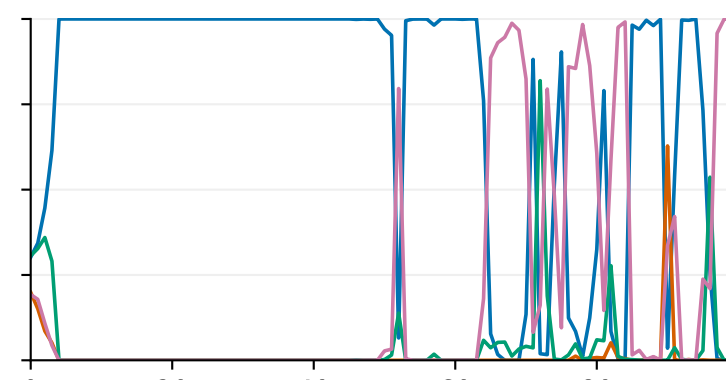
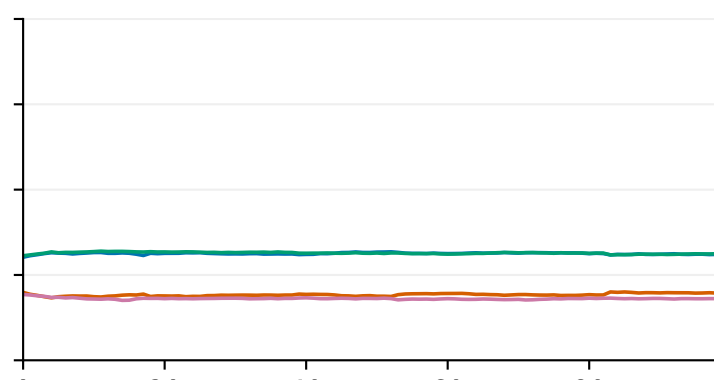
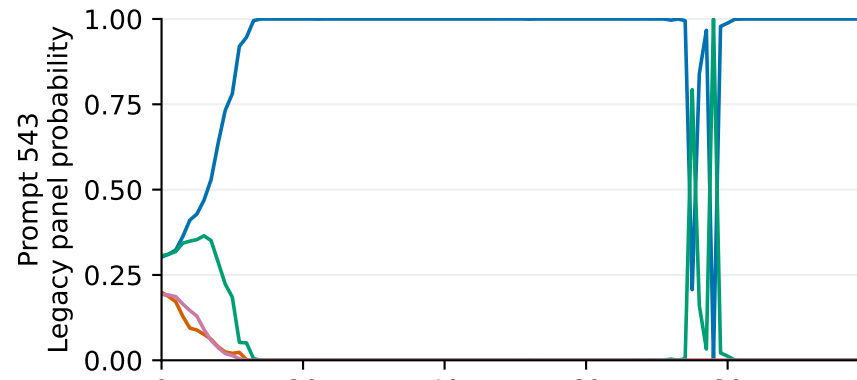
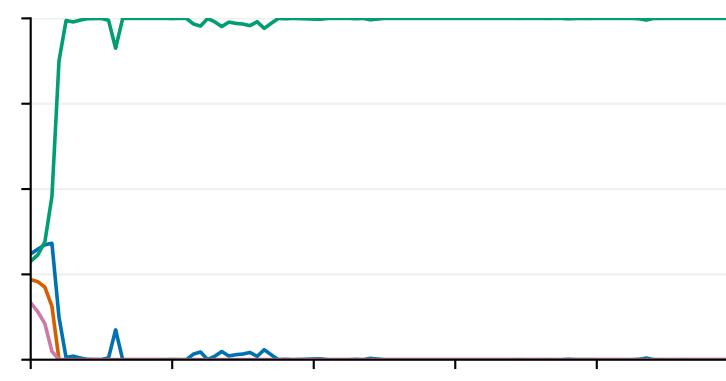
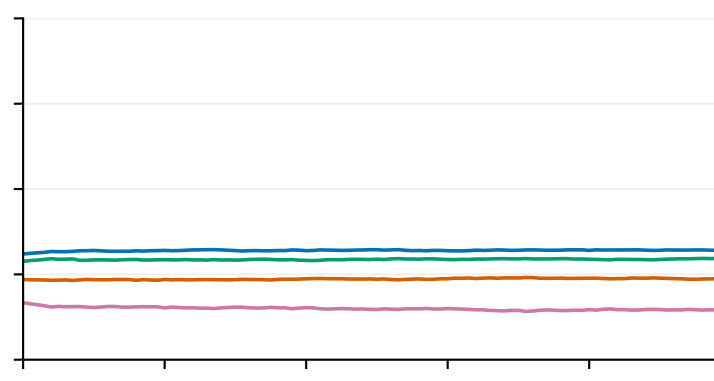
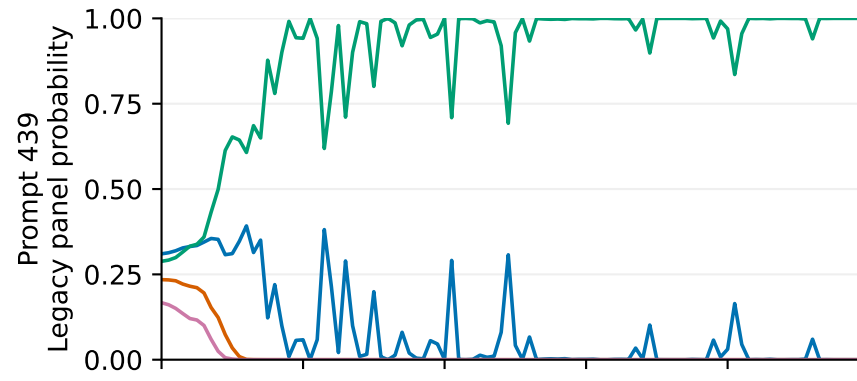
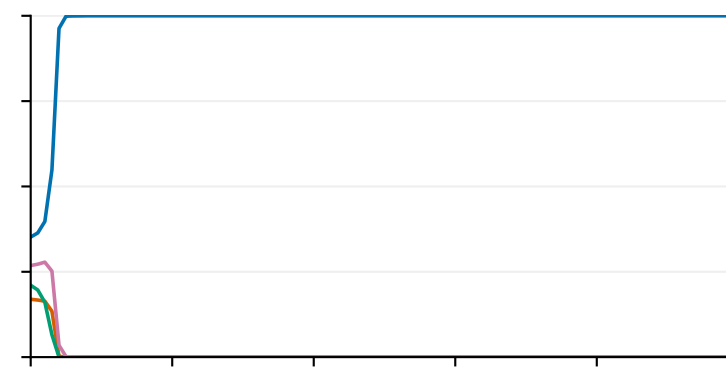
$\alpha=0.1, \lambda=0.1, \beta=0.01$



$\alpha=0.5, \lambda=0.5, \beta=5$



$\alpha=0.99, \lambda=0.8, \beta=1$



Outer iteration (pre-update)

Outer iteration (pre-update)

Outer iteration (pre-update)

Response 1 Response 2 Response 3 Response 4